

SUDO CODE - Week 6: Survey on ANN, RNN, LSTM, and GRU

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Abstract

This survey provides a detailed analysis of four major neural network architectures widely used in Natural Language Processing (NLP) before the emergence of Transformer models: Artificial Neural Networks (ANN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU). For each architecture, we first provide a step-by-step simplified explanation without unexplained jargon, followed by a deeper analysis of its characteristics, advantages, disadvantages, relationships with other models, and specific applications in NLP. The goal is to build a comparative understanding of how each architecture evolved to address the limitations of the previous one and to situate them within the broader development of sequence modeling in NLP.

1 Artificial Neural Networks (ANN)

Definition

- ANN is like a machine built from many tiny boxes called **neurons**.
- Each neuron:
 - Takes a number in.
 - Performs simple math (addition, multiplication).
 - Passes the result to the next neuron.
- Neurons are grouped into layers:
 - Input layer: receives raw data (e.g., pixels, words converted into numbers).
 - Hidden layers: detect useful patterns.
 - Output layer: produces the final answer (e.g., “spam” vs. “not spam”).
- Neurons learn by adjusting **weights**, like tuning knobs on a radio.

Properties and Analysis

- **Purpose:** Designed to learn patterns in data without handcrafting rules.
- **Strengths:** Good for classification, regression, and image tasks.
- **Weaknesses:** Cannot understand order or time. The sequence “dog bites man” and “man bites dog” are treated the same.
- **Relation:** Basis for more advanced sequence models like RNNs.

Applications in NLP

- Spam filtering.
- Text classification with bag-of-words (ignores word order).
- Topic categorization.

2 Recurrent Neural Networks (RNN)

Definition

- RNN is like ANN but with **short-term memory**.
- Imagine reading a story word by word while remembering what came before.
- At each step, RNN updates a **hidden state** that carries information forward.

Properties and Analysis

- **Purpose:** Solve ANN's weakness by remembering order in data.
- **Strengths:** Handles sequential data (text, audio, time series).
- **Weaknesses:** Suffers from the **vanishing gradient problem**, making it hard to learn long-term relationships.
- **Relation:** Natural extension of ANN to sequence tasks.

Applications in NLP

- Language modeling (predict next word).
- Early machine translation.
- Speech recognition.

3 Long Short-Term Memory (LSTM)

Definition

- LSTM is a smarter RNN with a **notebook system**.
- It uses three gates to manage memory:
 - **Forget gate:** decides what to erase.
 - **Input gate:** decides what to add.
 - **Output gate:** decides what to use right now.
- This allows it to remember important things for a long time.

Properties and Analysis

- **Purpose:** Solve RNN's vanishing gradient problem.
- **Strengths:** Can remember long-term dependencies (e.g., "if" at the start linked to "then" later).
- **Weaknesses:** Complex, requires more computation, slower to train.
- **Relation:** Extension of RNN with memory gates; GRU later simplified this design.

Applications in NLP

- Machine translation (Google Translate pre-Transformers).
- Text generation (story writing, chatbots).
- Named Entity Recognition (NER).
- Sentiment analysis with long sentences.

4 Gated Recurrent Unit (GRU)

Definition

- GRU is a simpler version of LSTM.
- It uses only two gates:
 - **Update gate:** decides what past info to keep.
 - **Reset gate:** decides what past info to forget.
- Easier to train but still effective.

Properties and Analysis

- **Purpose:** Provide LSTM's benefits with less complexity.
- **Strengths:** Faster, uses fewer resources, good with small datasets.
- **Weaknesses:** Sometimes less powerful for very long sequences compared to LSTM.
- **Relation:** Simplified variant of LSTM; part of RNN family.

Applications in NLP

- Sentiment analysis.
- Question answering.
- Machine translation on resource-constrained devices.

5 Comparison and Relationships

Key Differences

- **ANN:** Learns patterns but ignores order.
- **RNN:** Adds sequence memory but fails for long dependencies.
- **LSTM:** Fixes memory issue with gates, but heavier.
- **GRU:** Simpler and faster than LSTM, slightly less expressive.

Evolution Path

- ANN $\rightarrow$ RNN $\rightarrow$ LSTM/GRU.
- Each model fixes limitations of the previous one.
- All paved the way for Transformer models, which now dominate NLP.

Use in NLP Tasks

- ANN: simple classification (spam detection).
- RNN: sequence modeling (language modeling).
- LSTM: long context tasks (translation, summarization).
- GRU: efficient tasks where speed and fewer resources matter.

6 References

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